

CIS 635: Data Mining and Knowledge Discovery

Team Project Guidelines and Requirements

Fall 2026

1 Overview

Each student will complete a data mining and knowledge discovery (KDD) project, working in a team of up to two students. The project applies the concepts, tools, and techniques covered in this course to a real, multi-dataset problem, following the KDD process from data understanding through evaluation and communication of results.

2 Team Formation

- Teams may consist of one or two students, with two-student teams preferred. Students select their own teammates; the instructor can help match students who need a partner upon request.
- Team composition must be finalized and reported to the instructor before the Project Specification Document is submitted in Week 4.

3 Project Timeline and Deliverables

The table below summarizes all required project milestones.

When	Deliverable / Milestone	Notes
Week 4	Project Specification Document due	See Section 5 for required content; sample provided in the Appendix
By Week 5	Development tool validated (if not using Colab)	See Section 4
Weeks 5-7	Progress Check-In #1	Brief meeting with instructor; see Section 6
Weeks 9-11	Progress Check-In #2	Brief meeting with instructor; see Section 6
Weeks 13-14	Progress Check-In #3 and Final Presentation	See Sections 6-7
Week 14	Final Project Report due	See Section 7

4 Development Environment

- Google Colab is the primary/default development environment for all teams (Python-based notebooks, free GPU access, no local setup required).

- Teams may propose an alternative environment or toolset (e.g., Jupyter locally, RStudio, a specific cloud platform), provided it is submitted for instructor review and validated no later than Week 5.
- Tool-change requests should briefly state the proposed tool and why it is needed (e.g., a package or dataset-size requirement Colab cannot support).
- Until an alternative is approved in writing, teams should proceed as if Colab is required.

5 Dataset Requirements

Each project must use **at least two genuinely inter-related datasets**. The datasets must share a meaningful connection—such as a common key, entity, geography, time period, or subject—that allows them to be **joined or combined into a single analysis**. Datasets that are merely analyzed side by side do not meet this requirement. **Any exception to this requirement must be approved by the instructor.**

5.1 What counts as “inter-related”

The datasets must be linkable through one or more of the following, and the project must actually use that link (e.g., via a join, merge, or shared grouping variable):

- A shared identifier or key (e.g., patient ID, sample ID, customer ID, product ID).
- A shared geographic dimension (e.g., county, state, ZIP code, region).
- A shared temporal dimension (e.g., matching date ranges or reporting periods).
- A logical/contextual relationship where one dataset provides features, labels, or context for records in the other (e.g., clinical outcomes linked to molecular data for the same subjects).

5.2 Examples

Acceptable (genuinely related):

- A gene-expression dataset and a clinical/phenotype metadata table for the same set of samples (shared sample ID).
- A patient encounter or clinical dataset joined with a county-level public-health or social-determinants dataset (shared geography).
- A clinical trials registry joined with an adverse-event or outcomes dataset (shared trial ID).
- Customer transaction data joined with customer demographic or survey data (shared customer ID).

Not acceptable (independent datasets with no real link):

- Two unrelated public datasets from different domains that are each analyzed on their own with no shared key, join, or combined analysis.

5.3 Dataset Ethics and Privacy

- All datasets must be publicly available, properly licensed, or otherwise approved in advance for academic use.
- Datasets must not contain personally identifiable information (PII) or protected health information (PHI) beyond what is already de-identified in a public, approved repository (e.g., de-identified EHR extracts, public gene-expression archives, government open-data portals).
- No IRB approval is required for using existing, publicly available, de-identified datasets for this course project.
- If a team wants to use a restricted or identifiable dataset (e.g., data requiring a data-use agreement or containing PHI), they must consult the instructor before proceeding; such data is generally not appropriate for a project on this timeline.
- Every dataset used must be cited in the specification document and final report, including its source and license/terms of use.

6 Progress Check-Ins

Each team meets briefly with the instructor three times during the semester to report on progress, ask questions, and get feedback. These are short, informal check-ins rather than written reports.

Check-in	Window	Focus
#1	Weeks 5-7	Confirmed datasets and relationship, initial data exploration/cleaning, refined scope
#2	Weeks 9-11	Feature engineering / preprocessing complete, initial modeling or analysis results
#3	Weeks 13-14	Final results, remaining work, and presentation readiness (paired with the final presentation)

- Teams sign up for a meeting slot with the instructor within each check-in window (in person or virtual).
- Come prepared with a one-paragraph status update and any specific questions or blockers.

7 Final Report and Presentation (Week 14)

7.1 Final Project Report

Due Week 14. Suggested structure, following the KDD process:

1. Introduction and problem motivation
2. Data description (both/all datasets, sources, sizes, and how they relate)
3. Data preprocessing and integration (cleaning, joining, feature engineering)
4. Methodology (techniques applied and why they fit the problem)

5. Results
6. Discussion and limitations
7. Conclusion and possible future work
8. References and dataset citations
9. Appendix: link to the Colab notebook (or other approved environment) with runnable code

7.2 Final Presentation

- Delivered during the Week 13-14 window alongside Check-in #3.
- Suggested length: 10-15 minutes per team, including time for questions.
- Should summarize the problem, datasets and their relationship, methods, key results, and lessons learned.

8 Grading

This project counts for **30% of the course grade**. The grading breakdown is as follows:

Component	Weight	of Course Grade
Project Specification Document (Week 4)	3%	of 30%
Progress Check-Ins (3, combined)	7%	of 30%
Final Project Report	10%	of 30%
Final Presentation	10%	of 30%
Total	30%	of course grade

Appendix: Sample Project Specification Document

The following is a complete, worked example of a feasible project specification for a team of up to two students, scoped to 6-8 weeks of implementation. It illustrates the required content and level of detail; teams should replace it with their own topic, not reuse it.

Title

Predicting Breast Cancer Molecular Subtype from Linked Gene-Expression and Clinical Metadata

Team

[Student A Name], [Student B Name]

1. Problem Statement and Motivation

Molecular subtyping of breast cancer (e.g., by estrogen-receptor, progesterone-receptor, and HER2 status, or intrinsic subtypes such as Luminal A/B, HER2-enriched, Basal-like) guides treatment decisions. Gene-expression profiling can reveal these subtypes directly from tumor tissue. This project asks whether a small set of expression-derived features can predict clinically annotated subtype, and whether unsupervised clustering of expression profiles recovers the same groupings as the clinical labels: a compact but realistic example of linking molecular and clinical data in a KDD workflow.

2. Datasets

Dataset 1 (Gene expression matrix): A public gene-expression series for breast cancer from the NCBI Gene Expression Omnibus (GEO), containing normalized expression values for thousands of genes across roughly 100-300 tumor samples, indexed by sample accession (GSM ID). Format: tab-delimited/CSV series matrix. License: public, NIH-hosted, free for research and educational use.

Dataset 2 (Clinical/phenotype metadata): The accompanying sample-characteristics table for the same GEO series, reporting per-sample fields such as age, tumor stage, ER/PR/HER2 status, and (where available) intrinsic subtype or survival outcome, indexed by the same GSM sample accession. Format: CSV. License: same public GEO terms.

Before finalizing the dataset choice, confirm the exact fields available in the specific GEO series selected (not every series reports every clinical field) and adjust the target label accordingly (e.g., ER status if intrinsic subtype is not annotated).

3. Dataset Relationship

Both datasets share the GEO sample accession (GSM ID) as a common key. Joining them produces a single sample-level table combining molecular features (gene expression) with clinical labels for the same tumor samples. The expression data alone has no outcome to predict, and the clinical data alone has no molecular features, so the analysis depends on the join.

4. Proposed KDD Methods

- Preprocessing: handle missing values, normalize/scale expression values, encode clinical fields.

- Feature selection / dimensionality reduction: select the most variable or differentially expressed genes; apply PCA for visualization and noise reduction.
- Supervised learning: classification (e.g., logistic regression, random forest) to predict receptor status or subtype from expression features.
- Unsupervised learning: k-means or hierarchical clustering on expression profiles, compared against known clinical subtypes.
- Evaluation: accuracy/F1/ROC-AUC for classification; silhouette score and cross-tabulation with clinical labels for clustering.

5. Feasibility and Scope Justification (6-8 weeks)

The data is public and immediately downloadable (no collection needed). Sample size (approximately 100-300) and a focused set of engineered features keep preprocessing and modeling tractable in Colab without specialized compute. Combining one classification task and one clustering task is a standard, well-scoped analysis for two students in this timeframe; the plan avoids open-ended data collection or infrastructure work.

6. Preliminary Work Plan

Week	Planned work
1	Acquire both datasets; initial cleaning and join on sample ID
2	Exploratory data analysis; handle missing data; normalize expression values
3	Feature selection (top variable/differentially expressed genes); PCA
4	Baseline classification model(s)
5	Clustering analysis; compare clusters to clinical labels
6	Model tuning and evaluation; compare approaches
7	Draft final report; prepare presentation
8 (buffer)	Finalize report, polish notebook, rehearse presentation

7. Tools and Environment

Google Colab (Python: pandas, NumPy, scikit-learn, seaborn/matplotlib; optionally GEOparse for retrieving GEO records).

8. Expected Challenges and Risks

- High dimensionality (thousands of genes) relative to sample count: mitigated via feature selection/PCA.
- Inconsistent or missing clinical annotation across samples: mitigated by choosing a well annotated GEO series and documenting any exclusions.
- Possible batch effects if samples are pooled from multiple sub-series: mitigated by preferring a single, consistently processed series.

- Class imbalance among subtypes: addressed via appropriate evaluation metrics (F1, balanced accuracy) rather than raw accuracy alone.